

Project Report
On
Computer Networking: Concepts
(CSE3751)

CENTRALISED FILE SHARING SYSTEM
WITH DHCP & FTP SERVER



Submitted by:

Name1: SURAJIT SAHOO

Regd. No.: 2341019165

Name2: SATYAJIT SETHY

Regd. No.: 2341019390

Name3: E. SAILAJA

Regd. No.: 2341019290

Name4: TRIBHUWAN SINGH

Regd. No.: 2341019538

B. Tech. CSE(AI & ML) 5th Semester (Section 23412C3)

INSTITUTE OF TECHNICAL EDUCATION AND RESEARCH
(FACULTY OF ENGINEERING)
SIKSHA 'O' ANUSANDHAN (DEEMED TO BE UNIVERSITY),
BHUBANESWAR, ODISHA

We, the undersigned students of B. Tech. of **COMPUTER SCIENCE & ENGINEERING WITH SPECIALISATION IN AI/ML** Department hereby declare that we own the full responsibility for the information, results etc. provided in this PROJECT titled “**CENTRALISED FILE SHARING SYSTEM WITH DHCP AND FTP SERVER**” submitted to **Siksha ‘O’ Anusandhan (Deemed to be University), Bhubaneswar** for the partial fulfillment of the subject **Computer Networking: Concepts (CSE 3751)**. We have taken care in all respect to honor the intellectual property right and have acknowledged the contribution of others for using them in academic purpose and further declare that in case of any violation of intellectual property right or copyright we, as the candidate(s), will be fully responsible for the same.

Name1: SURAJIT SAHOO Regd. No.: 2341019165

Name2: SATYAJIT SETHY Regd. No.: 2341019390

Name3: E. SAILAJA Regd. No.: 2341019290

Name4: TRIBHUWAN SINGH Regd. No.: 2341019538

Date: 06.01.26

Place: ITER COLLEGE, BHUBANESWAR

Abstract

This project presents the design and implementation of a centralised file-sharing system for a single-office organisation using Cisco Packet Tracer. The network consists of three departments - HR, IT, and Finance, each placed in separate IP subnets to ensure logical departmental isolation. A centralized FTP server hosted in the IT department enables secure file sharing across departments with user authentication. Dynamic Host Configuration Protocol (DHCP) is implemented to automatically assign IP addresses to all client devices, reducing manual configuration errors. Inter-department communication is achieved using static routing through a central router. The project demonstrates successful IP assignment, routing, and file transfer operations, validating the effectiveness of the proposed network design. This implementation provides a scalable, organized, and efficient network solution suitable for small office environments.

Contents

Serial No.	Chapter No.	Title of the Chapter	Page No.
1.	1	Introduction	5
2.	2	Problem Statement	6
3.	3	Methodology	7
4.	4	Results and interpretation	9
5.	5	Conclusion	11
6.	6	References	12

1. Introduction

Modern organisations require efficient mechanisms for sharing files and managing network resources across multiple departments. Manual IP configuration and direct file exchange between systems often lead to mismanagement, security risks, and scalability issues.

This project focuses on designing a centralised file-sharing system where all departments are connected through a structured network. Each department is logically separated using distinct subnets, while centralised services such as DHCP and FTP are hosted within the IT department. Cisco Packet Tracer is used to simulate and validate network behaviour. The project aims to demonstrate automated IP address allocation, controlled inter-department communication, and secure file sharing using FTP.

2. Problem Statement

The objective of this project is to design and implement a centralised file-sharing system using Cisco Packet Tracer. The organisation consists of HR, IT, and Finance departments, each operating in separate subnets. The IT department hosts an FTP server to facilitate centralised document sharing.

All client systems must receive IP addresses dynamically through a DHCP server. Communication between departments must be enabled using static routing. Additionally, logical departmental isolation must be maintained to ensure organised network segmentation.

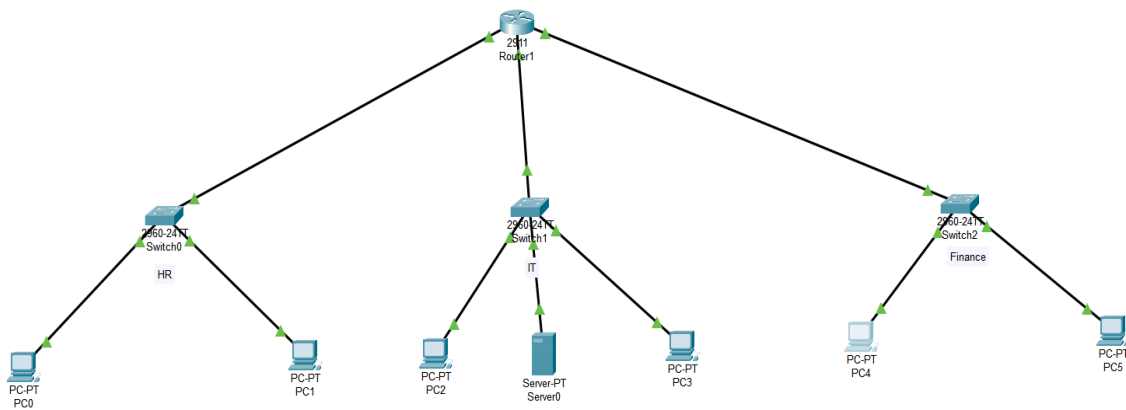
The implementation should satisfy the following constraints:

- Use of Cisco Packet Tracer simulation environment
- Centralised DHCP and FTP services
- No VLAN configuration (Layer-3 subnet-based isolation)

3.Methodology

3.1 Network Topology Design

The network topology consists of one central router, three access switches, six client PCs, and one server. Each department—HR, IT, and Finance—is connected to the router via a dedicated switch and operates on a unique IP subnet. The server is placed in the IT department subnet and provides DHCP and FTP services to the entire network. Copper straight-through cables are used for all device connections.



3.2 IP Addressing Scheme

Department	Network Address	Gateway
HR	192.168.10.0 /24	192.168.10.1
IT	192.168.20.0 /24	192.168.20.1
Finance	192.168.30.0 /24	192.168.30.1

The router interfaces are assigned the first usable IP address in each subnet, while the server is assigned a static IP address of 192.168.20.2.

3.3 Device Configuration

(a) Router Configuration

The router is configured with static IP addresses on each interface corresponding to the departmental subnets. Since all networks are directly connected, static routing is inherently achieved.

CLI command :

```
interface g0/0
ip address 192.168.10.1 255.255.255.0
no shutdown
```

(b) DHCP Configuration

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
Finance	192.168.30.1	8.8.8.8	192.168.3...	255.255.2...	50	0.0.0.0	0.0.0.0
IT	192.168.20.1	8.8.8.8	192.168.2...	255.255.2...	50	0.0.0.0	0.0.0.0
HR	192.168.10.1	8.8.8.8	192.168.1...	255.255.2...	50	0.0.0.0	0.0.0.0

A centralised DHCP server is configured on the IT department server. Separate DHCP pools are created for HR, IT, and Finance departments. Since the DHCP server resides in a different subnet, DHCP relay is enabled on the router using the ip helper-address command.

(c) FTP Configuration

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	finuser	fin123	RWL
3	hruser	hr123	RWL
4	ituser	it123	RWL

The FTP service is enabled on the server with authentication. User accounts are created for each department with read and write permissions to allow secure file upload and download operations.

4. Results & Interpretation

4.1 DHCP Verification

```
Router>show ip interface brief
```

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	192.168.10.1	YES	manual	up	up
GigabitEthernet0/1	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2	192.168.30.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

All client PCs successfully received IP addresses dynamically from the DHCP server. Each PC was assigned an IP address corresponding to its departmental subnet along with the correct default gateway.

4.2 Inter-Department Communication Verification

```
C:\>ping 192.168.20.2
```

Pinging 192.168.20.2 with 32 bytes of data:

```
Reply from 192.168.20.2: bytes=32 time=12ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
Reply from 192.168.20.2: bytes=32 time<1ms TTL=127
```

Ping statistics for 192.168.20.2:

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms
```

Connectivity between departments was verified using the ping command. Successful ICMP replies confirmed proper routing between different subnets.

4.3 FTP File Transfer Verification

```
C:\>ftp 192.168.20.2
Trying to connect...192.168.20.2
Connected to 192.168.20.2
220- Welcome to PT Ftp server
Username:hruser
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>put test.txt

Writing file test.txt to 192.168.20.2:
File transfer in progress...

[Transfer complete - 15 bytes]

15 bytes copied in 0.077 secs (194 bytes/sec)
```

```
C:\>ftp 192.168.20.2
Trying to connect...192.168.20.2
Connected to 192.168.20.2
220- Welcome to PT Ftp server
Username:finuser
331- Username ok, need password
Password:
230- Logged in
(passive mode On)
ftp>get test.txt

Reading file test.txt from 192.168.20.2:
File transfer in progress...

[Transfer complete - 15 bytes]

15 bytes copied in 0 secs
```

FTP connectivity was verified by uploading a file from an HR department PC to the centralised FTP server and subsequently downloading the same file from a Finance department PC. This confirms successful centralised file sharing across departments.

5. Conclusion

This project successfully demonstrates the design and implementation of a centralised file-sharing system using Cisco Packet Tracer for a single-office organisational setup. The network was structured to include HR, IT, and Finance departments, each operating within its own IP subnet, thereby achieving logical departmental isolation. This segmentation improves network organisation, security, and manageability while allowing controlled inter-department communication.

A centralised DHCP server was implemented to dynamically assign IP addresses to all client systems, eliminating the need for manual configuration and reducing the possibility of IP conflicts. Since the DHCP server was located in a different subnet, a DHCP relay was configured on the router to ensure seamless IP allocation across all departments. Inter-subnet communication was enabled using static routing through a central router, ensuring reliable data transfer between departments.

Additionally, a centralised FTP server hosted in the IT department enabled secure and efficient file sharing across the organisation. Authentication-based access control ensured that only authorised users could upload and download files. Successful file transfer tests validated the functionality of the FTP service and the overall network connectivity.

Overall, the project meets all the stated objectives and provides a scalable, structured, and efficient network design. The implementation reflects real-world networking practices and serves as a practical model for centralised resource management in small to medium-sized organisational environments.

References

- [1] CompTIA Network+ N10-008 Certification Guide by Glen D. Singh, 2nd Edition, Packt publication.
- [2] G. D. Singh, “CompTIA Network+ N10-008 Certification Guide,” 2nd ed., Packt Publishing, 2022.
- [3] Cisco Networking Academy, “Introduction to Networks,” Cisco Systems.